

Data Warehouse

pros:

- Business Intelligence
- Analytics
- Structured & clean data
- Predefined schemas

Cons:

- No support for semi or unstructured data
- Inflexible schemas
- Struggled with volume and velocity, variety upticks
- Long processing time

2000s -> BIG DATA

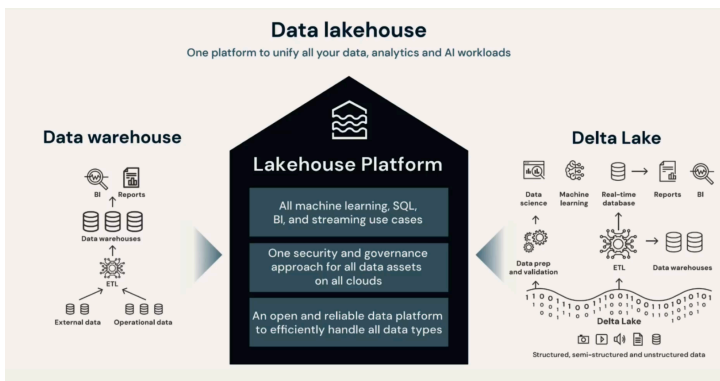
Data Lakes

Pros:

- Structured, semi or unstructured simultaneously stored in data lake
- Multiple data types could be stored
- Data created from different sources could be streamed quickly, cheaply in low cost cloud object storage
- Support for AI and Machine Learning

Cons:

- No ACID transactional support
- Lack of schema enforcement
- Lack of integration with a data catalog
- Poor data reliability (due to the various formats)
- Slow analysis performance (due to large volume data)
- Data governance concerns (security and privacy)
- Data warehouses still needed (store same information twice)



Data Lakehouse combines data warehouse and data lake.
It's the modernized version of data warehouse.

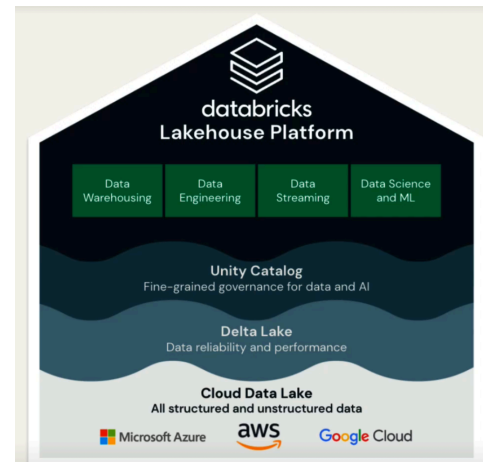
- Built on a data lake

Key features:

- Transaction support
- Schema enforcement and governance
- Data governance
- BI support
- Decoupled storage from compute
- Open storage format
- Support for diverse data types (structured, semi on unstructured)
- Support for diverse workloads
- End to end streaming

Databricks Platform:

- First lakehouse platform in the cloud
- Unified platform for data and AI data for all data types
- Open with single security and governance model
- Cloud agnostic, so data can be governed wherever it's stored
- On the Databricks Lakehouse Platform there are the following features:
 - Data reliability and performance in Data Lake Foundation
 - Governance for data and AI in Unity Catalog
 - Persona-based use cases
 - Instant serverless compute (Databricks manages the compute layer on behalf of the customer)



Data lakes good for great amount of raw data, but bad for quality of data. Bad performances compared to data warehouse (ineffective partitioning and too many small files). Small files problem = bad query performances. Problemi risolti con

Delta Lake + Photon

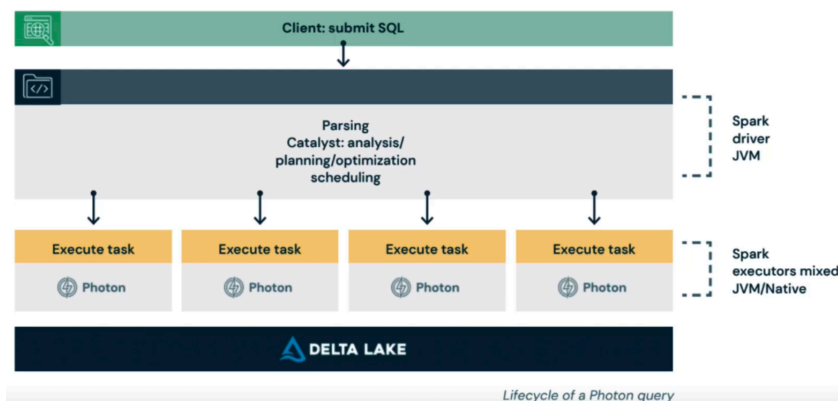
Delta Lake

- Data storage format built for the lake house and
- Acid transaction guarantees
- Scalable data and metadata Handling
- Audit history and time travel (dai transaction log)
- Schema enforcement and evolution
- Support for deletes updates and merges
- Unified streaming and batch data processing
- Compatible with Apache Spark
- Uses **delta Tables** (basate su Apache Parquet format for semi and unstructured data)
- Integrata coi + famosi tool di ingestion, query, store

Photon

- Execution engine that provides the same performances of a data warehouse
- Query engine
- Infrastructure cost savings
- Compatible with Spark APIs
- Increased speed for data ingestion, ETL, streaming, data science, interactive queries
- IoT use cases
- Load data into Delta Lake and Parquet
- Accelerates all data analytics workloads, not only SQL queries

Photon in the Databricks Lakehouse Platform

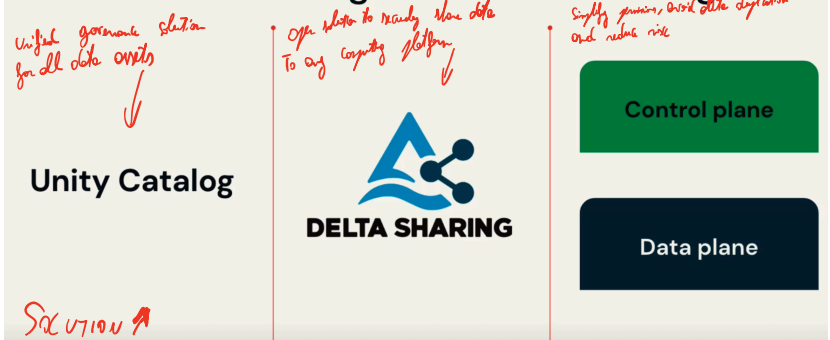


Data Governance

Challenges:

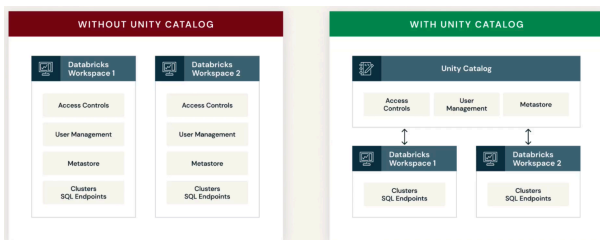
- Diversity of data and AI assets
- Using incompatible data platforms
- Multi-cloud
- Fragmented tool usage for data governance.

How the Databricks Lakehouse Platform solves data and AI governance challenges



Unity Catalog

- Common data governance model based on ANSI SQL (asset in any cloud)
- One model to discover, access, share data
- Centralised access control, user management, metastore
- Restrict access to some rows and columns (dice chi ha avuto accesso a cosa e quando)
- Detailed logging



Control Plane

- Managed backend services that Databricks provides
- In Databricks' own cloud account
- Aligned with customer services (GCP, AWS, Azure)
- Manages notebooks, configurations, clusters, logs

Data Plane

- Where data is processed
- Compute resources run inside the business owner's own cloud account
- Data stays where it is
- Communication to and from the control plane is encrypted

User identity and access

- Table ACLs feature
- IAM instance profiles
- Securely stored access key
- Secrets API

Delta Sharing

- Open source solution to share live data from your lake house to any computing platform securely
- Recipients don't have to be on the same cloud
- Share live data without copying it
- Centralised administration and governance
- Marketplace for data products
- Privacy safe
- Simple REST protocol

	Proprietary vendor solutions	SFTP	Cloud object store	Delta Sharing
Secure	✓	✓	✓	✓
Cheap		✓	✓	✓
Vendor agnostic		✓		✓
Multicloud		✓		✓
Open source		✓		✓
Table/DataFrame abstraction	✓			✓
Live data	✓			✓
Predicate pushdown	✓			✓
Object store bandwidth			✓	✓
Zero compute cost			✓	✓
Scalability		✓		✓

Serverless Data Plane

- Nel classico data plane le compute resources sono nel business's cloud storage account ma ci sono dei problemi (cluster creation complicated, environment startup slow, since it's in the business cloud account there are limitations, over provisioning of resources ecc)
- Serverless data plane risolve questo ed è disponibile solo per Databricks SQL e risolve tutti i problemi precedenti
- Completely managed by Databricks and it's on demand
- Elastic - scale up or down as needed
- 3 layers of isolation: the container hosting runtime, the VM hosting the container, the virtual network for the workspace. Each part is isolated

Metastore: represents the metadata (offer improved security and auditing capabilities)

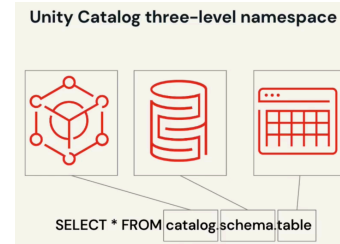
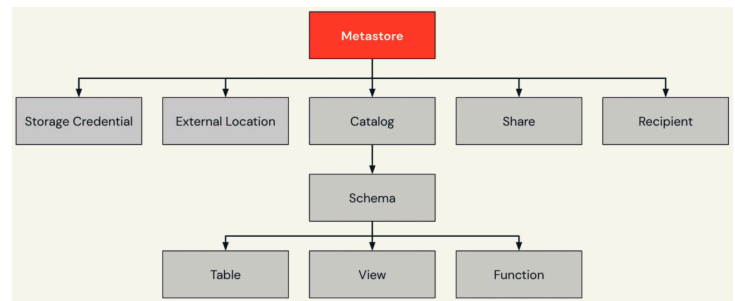
Catalog: Top container for data objects in unity catalog. Un metastore può avere + catalogs. È il primo dei 3 livelli del namespace usato dai data analyst.

Schema: Ugualo al SQL cioè container per data assets come tables e views. È il 2 punto del namespace. Il catalog può contenere + schema.

Table: SQL relations consisting of an ordered list of columns. 2 variazioni, sono definite da: metadata (comments, tags, list of columns) e data types. 2 tipi di tabelle, Managed ed External, le prime sono salvate in metastore managed stores location, le seconde in un external storage location.

Views: sono stored queries, executed when you query the view. Read only.

UDF: Encapsulate custom functionality into a function evoked within queries.



Why data is a valuable asset?

Data engineering team focuses around ingesting, transforming, orchestrating data to different teams for day-to-day insights, innovation, and tasks.

Challenges:

- Complex data ingestion methods
- Support for data engineering principles (agile, isolated development and production environments, CI/CD, version control transformations)
- 3rd party orchestration tools (+ overhead, - reliability)
- Pipeline and architecture tuning.
- Inconsistencies between data warehouse and data lake providers

Databricks offre:

A unified data platform with managed data ingestion, schema detection, enforcement, and evolution, paired with declarative, auto-scaling data flow integrated with a lake house native orchestrator that supports all kinds of workflows.

- Data engineers can focus only on quality and reliability. Because:
 - Easy data ingestion
 - Automated ETL pipelines
 - Data quality checks
 - Batch and streaming tunings
 - Automatic recovery
 - Data pipeline observability
 - Scheduling and orchestration

Data Ingestion:

- Automatic infer the schema and evolves as the data comes in
- Auto loader: data ingestion tool that processes new data as they arrive in the lakehouse cloud storage.
- COPY INTO SQL load data from a folder into delta lake table

Data transformation

- DLT Delta Live Tables: ETL framework that uses a simple declarative approach to build reliable data pipelines
 - Automatically scales the infrastructure so focus only on getting value from data
- **Databricks Workflows** Orchestrate any task across any platform.

Real time streaming data

- Generated by: transaction (POS, banking, etc), IoT, Interactions (web, social, emails)

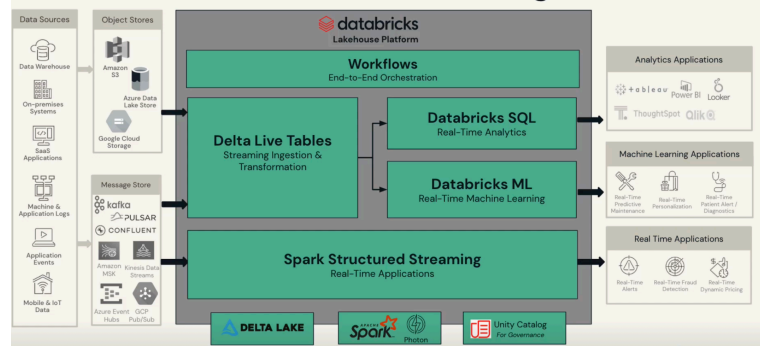
Use cases

- Real time analytics
- Real Time Machine learning
- Real time applications
- Retail - industrial automation - healthcare - financial institutions.

Why Databricks?

- Build streaming pipelines and applications faster
- Simplify operations with automation
- Unified governance for real time and historical data

Reference architecture for streaming use cases



Challenges to successful machine learning and AI endeavours

- Siloed and disparate data systems
- Complex experimentation environments
- Getting models to production
- Multiple tools available
- Experiments hard to track
- Reproducing results is difficult
- ML is hard to deploy

Databricks lakehouse platform provides a space for data scientists, ML engineers to derive innovative insights, build powerful predictive models etc.